Insurance CX: Claims & Sentiment Analysis

Bridging structured claims data with customer feedback to drive business insights and improve customer experience.

Made with GAMMA

Project Overview

The Challenge

Insurance companies rely on structured data but miss valuable insights hidden in customer feedback and unstructured data sources.

Our Solution

Combining insurance claims data with customer sentiment analysis to provide comprehensive views of business performance and customer experience.

Made with GAMMA

Key Challenges Addressed

High Claim Payouts

Managing claim payouts affecting profitability

Customer Visibility

Lack of visibility into customer satisfaction levels

Inefficient Processing

Slow and inefficient claim processing systems

Risk Identification

Difficulty identifying high-risk customers

Made with GAMMA

Project Objectives

01

Analyze Claims & Policy Performance

Evaluate claim frequency, amounts, and premium relationships

02

Identify Patterns

Detect trends in claim frequency and claim amount distributions

03

Assess Customer Sentiment

Perform sentiment analysis on customer feedback and reviews

04

Generate Insights

Provide data-driven recommendations for business improvement

Made with GAMMA

Dataset Overview

Insurance Dataset

- 10,000+ total records
- Customer information: Age, Gender
- Policy details: Type, Premium, Coverage
- Claims data: Amount, Status, Date

Feedback Dataset

- Customer reviews and feedback
- Sentiment analysis targets
- Service quality indicators
- Common issues and complaints

Made with GAMMA

Tools & Technologies

SQL Server

Data storage and querying for structured insurance data

Power BI

Interactive data visualization and dashboard creation

DAX

Calculated measures and KPIs for analysis

Excel/CSV

Data source files and initial data preparation

Made with GAMMA

Data Preprocessing

Date Conversion

Converted date columns to proper datetime format

Missing Values

Handled missing ClaimDate as no-claim cases

Data Quality

Removed duplicates and zero claim amounts

Segmentation

Created age groups for customer segmentation

Made with GAMMA

Feature Engineering

1

Age Groups

Categorized as Young Adult, Adult, Senior

2

Sentiment Category

Classified as Positive, Neutral, Negative

3

Sentiment Score

Numerical value for sentiment analysis

4

Satisfaction Score

Customer experience metric calculation

5

Urgency Level

Rated as Low, Medium, High priority

6

Feedback Categories

Claims, Customer Service, Pricing, etc.

Made with GAMMA

Key Insights

20%

Customer Issues

Customers reported service gaps requiring attention

2:1

Payout Ratio

Claim payouts significantly exceed premiums collected

Policy Impact

Certain policy types contribute more to claim frequency and require review

Service Delays

Claim processing inefficiencies and customer service delays are major concerns

Strong Correlation

Sentiment score and customer satisfaction show strong positive correlation

Made with GAMMA

Business Impact & Future Scope

Business Impact

- Identify high-risk customers and policies
- Improve customer experience through feedback analysis
- Support pricing and policy optimization
- Enable proactive issue resolution
- Enhance operational efficiency

Future Enhancements

- Machine learning for claim prediction
- Real-time dashboards with streaming data
- Enhanced NLP sentiment analysis
- Customer churn prediction models

Made with GAMMA